Quiz 9 Practice: Circles

MATH 130 · Wednesday, October 28 · Quiz 9 is in class on paper · Show your work for full credit

Name _____

Each section matches one kind of Quiz 9 question. Work the model with the class, then do the "your turn" problems on your own. Keep full calculator values until the last step, and round only the final answer as asked.

1 · Degrees to radians round to three decimal places

radians = degrees × π/180

Model Convert 141.5° to radians.

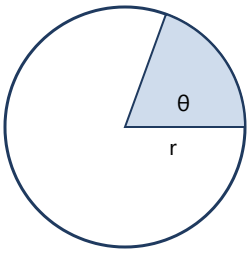
1 Convert 212.8° to radians.

2 Convert 47.25° to radians.

2 · Circumference, arc length, circle area, sector area round to one decimal place

$C = 2\pi r$ $s = (\theta/360)\cdot 2\pi r$

$A_{circle} = \pi r^2$ $A_{sector} = (\theta/360)\cdot \pi r^2$ (θ in degrees)

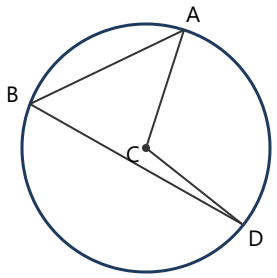


Model r = 4.2, θ = 110°. Find C, s, A_{circle}, and A_{sector}.

3 r = 7.5, θ = 48°. Find C, s, A_{circle}, and A_{sector}.

3 · Central and inscribed angles

Central angle = its intercepted arc. Inscribed angle = ½ × its intercepted arc.



Model C is the center and ∠ACD = 152°. Find (a) ∠ABD and (b) arc AD.

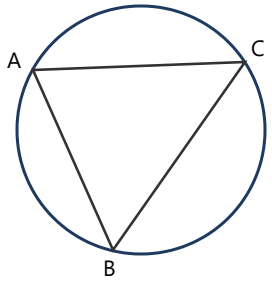
4 C is the center and ∠ACD = 118°. Find (a) ∠ABD and (b) arc AD.

5 Inscribed angle ∠ABD = 41°. Find (a) arc AD and (b) central angle ∠ACD.

4 · Inscribed triangles

Use the arc to find the inscribed angle that intercepts it, use the 180° triangle sum, then double an angle to get its arc.

Model Arc AB = 50° and ∠ABC = 72°. Find (a) ∠BAC and (b) arc BC.



6 Arc AB = 90° and ∠ABC = 55°. Find (a) ∠BAC and (b) arc BC.

7 Arc AB = 70° and ∠ABC = 60°. Find (a) ∠BAC and (b) arc BC.

5 · Volume word problems

Convert inches to feet first (3 in = 0.25 ft). Box: $V = lwh$. Cylinder: $V = \pi r^2 h$. "At least" means round the number of bags **up**.

Model A garden bed is 24 ft long and 5 ft wide. How many 1.5 ft^3 bags of mulch are needed to cover it with at least 4 inches of mulch?

8 A garden bed is 20 ft long and 7 ft wide. Mulch comes in 2 ft^3 bags. How many bags are needed for at least 3 inches of mulch?

9 A circular flower bed has radius 5 ft. How many 2 ft^3 bags are needed for at least 2 inches of mulch?

Check yourself: an arc can't be more than 360°, an inscribed angle is always smaller than its arc, and a sector's area is always smaller than the whole circle's.